

DSAI 490 – Assignment 1

Representation Learning with AE & VAE on Medical MNIST

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1 Architectures

Both the AE and VAE share a convolutional backbone on 64×64 grayscale inputs: three strided 3×3 conv blocks ($32 \rightarrow 64 \rightarrow 128$ channels) followed by a dense projection to a $D=16$ latent vector, and a mirrored transposed-conv decoder. The **AE** produces a deterministic code $z = E(x)$ and minimizes mean-squared error $\mathcal{L}_{\text{AE}} = \mathbb{E} \|x - D(z)\|^2$. The **VAE** head instead outputs $(\mu, \log \sigma^2)$; a sample is drawn via the reparameterization trick $z = \mu + \sigma \odot \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$. Its objective combines a per-pixel binary cross-entropy and the Gaussian KL to the prior:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q(z|x)}[-\log p(x|z)]}_{\text{recon}} + \beta \underbrace{\text{KL}(q(z|x) \parallel \mathcal{N}(0, I))}_{\text{KL}}.$$

We linearly anneal β from 0 to 1 over the first five epochs to avoid posterior collapse.

2 Setup

Medical MNIST (58k images, 6 classes) is streamed via `tf.data` with a fixed 80/10/10 train/val/test split per class. For each of the six anatomical regions we train one AE and one VAE (12 models); we additionally train one *global* AE and one *global* VAE on the union of all classes, used solely for the cross-class latent visualization – per-class encoders have unaligned latent axes, so stacking their codes would produce a misleading plot. All models use Adam ($\text{lr} = 10^{-3}$), batch size 128, and 20 epochs. See `notebook.ipynb` for the full pipeline.

3 Latent-space behavior

The AE’s latent space is compact but irregular: clusters are present but their relative positions reflect arbitrary encoder geometry. The VAE’s prior pulls codes toward $\mathcal{N}(0, I)$, yielding a more continuous manifold where interpolating between two codes decodes to a smooth morph between regions (see the latent-traversal figure in `figures/`).

4 Results

The AE achieves lower MSE on all six classes, as expected from its direct pixel-fidelity objective. SSIM tells a more nuanced story: the AE leads on AbdomenCT, CXR, ChestCT, and Hand, but the VAE

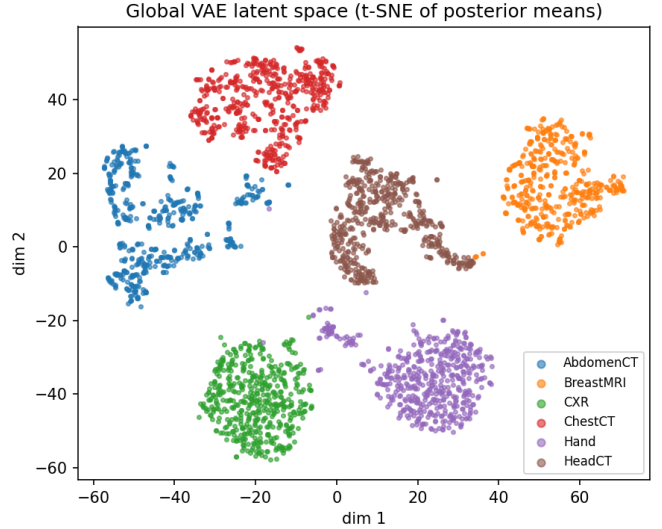


Figure 1: Global VAE latent space, t-SNE projection of posterior means on held-out data. Six anatomical regions form distinct but not isolated clusters, reflecting the smooth $\mathcal{N}(0, I)$ prior that is absent from the deterministic AE.

Class	AE MSE	AE SSIM	VAE MSE	VAE SSIM
AbdomenCT	0.0023	0.736	0.0038	0.637
BreastMRI	0.0060	0.751	0.0065	0.760
CXR	0.0081	0.709	0.0090	0.693
ChestCT	0.0023	0.643	0.0032	0.574
Hand	0.0082	0.673	0.0095	0.680
HeadCT	0.0103	0.706	0.0106	0.716

Table 1: Test-set reconstruction metrics on held-out test images.

matches or exceeds it on BreastMRI and HeadCT – classes with higher intra-class variability, where the VAE’s prior regularisation suppresses over-fitting artefacts. The VAE also supports *generation* from the prior (see `figures/vae_samples_global.png`) and smooth interpolation. For *denoising*, both models improve on the noisy input at $\sigma \in \{0.1, 0.2, 0.3\}$; the VAE degrades more gracefully at high σ because its prior-regularized decoder resists over-fitting to noise patterns.

5 Insights

KL warm-up was essential: without it, one or two classes drove the VAE into posterior collapse within the first epoch. Per-region reconstruction difficulty ranks, roughly, $\text{ChestCT} \approx \text{AbdomenCT} < \text{BreastMRI} < \text{CXR}$

$< \text{Hand} < \text{HeadCT}$ (by AE MSE), tracking intra-class variability. The cross-class t-SNE (Fig. 1) confirms that the global VAE’s latent captures anatomical-region identity without any class supervision – unsupervised representation learning is working as intended.

